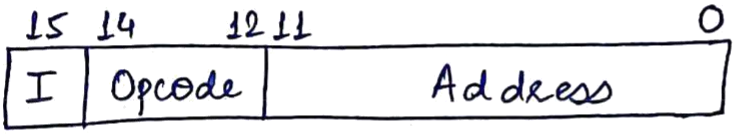
Computer Instructions

- The basic computer has three instruction code formats and each format has 16 bits.

a) Memory - Reference Instruction



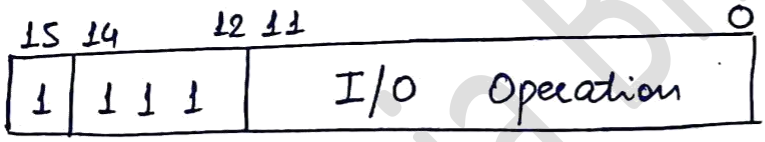
(Opcode = 000 - 110)

b) Register - Reference Instruction



(Opcode = 111, I = 0)

c) Input - Output Instruction



(Opcode = 111, I = 1)

Instruction Cycle

- A program residing in the memory unit of the computer consist of sequence of instructions.
- The program is executed in the computer by going through a cycle for each instruction.
- In the basic computer each instruction cycle consist of following phases:

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1. Fetch an instruction from memory.
2. Decode the instruction.
3. Read the effective address from memory if the instruction has indirect address.
4. Execute the instruction.

- Operations for Fetch and Decode

T_0 : AR ← PC

T_1 : IR ← M[AR], PC ← PC + 1

T_2 : D_0 ---- D_7 ← Decode IR (12-14)

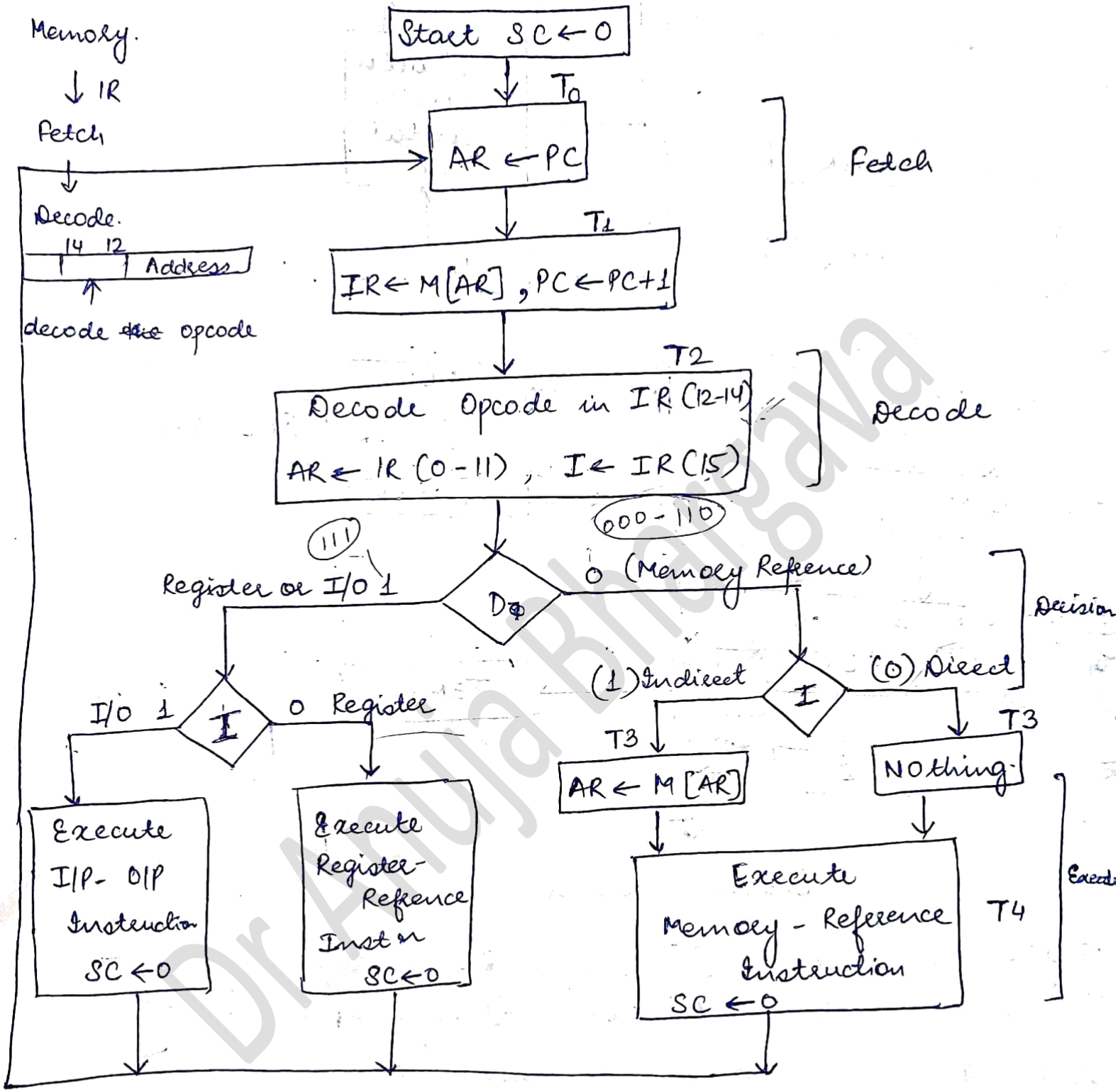
AR ← IR (0-11)

I ← IR (15)

- Initially, program counter PC is loaded with the address of the first instruction in the program.
- Place the content of PC into AR.
- The instruction is extracted in IR (instruction register from AR (Address Register)
- Last, decode the instruction register.
- The flow chart of instruction cycle is as shown

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Flow Chart of Instruction Cycle.



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Execution of Complete Instruction

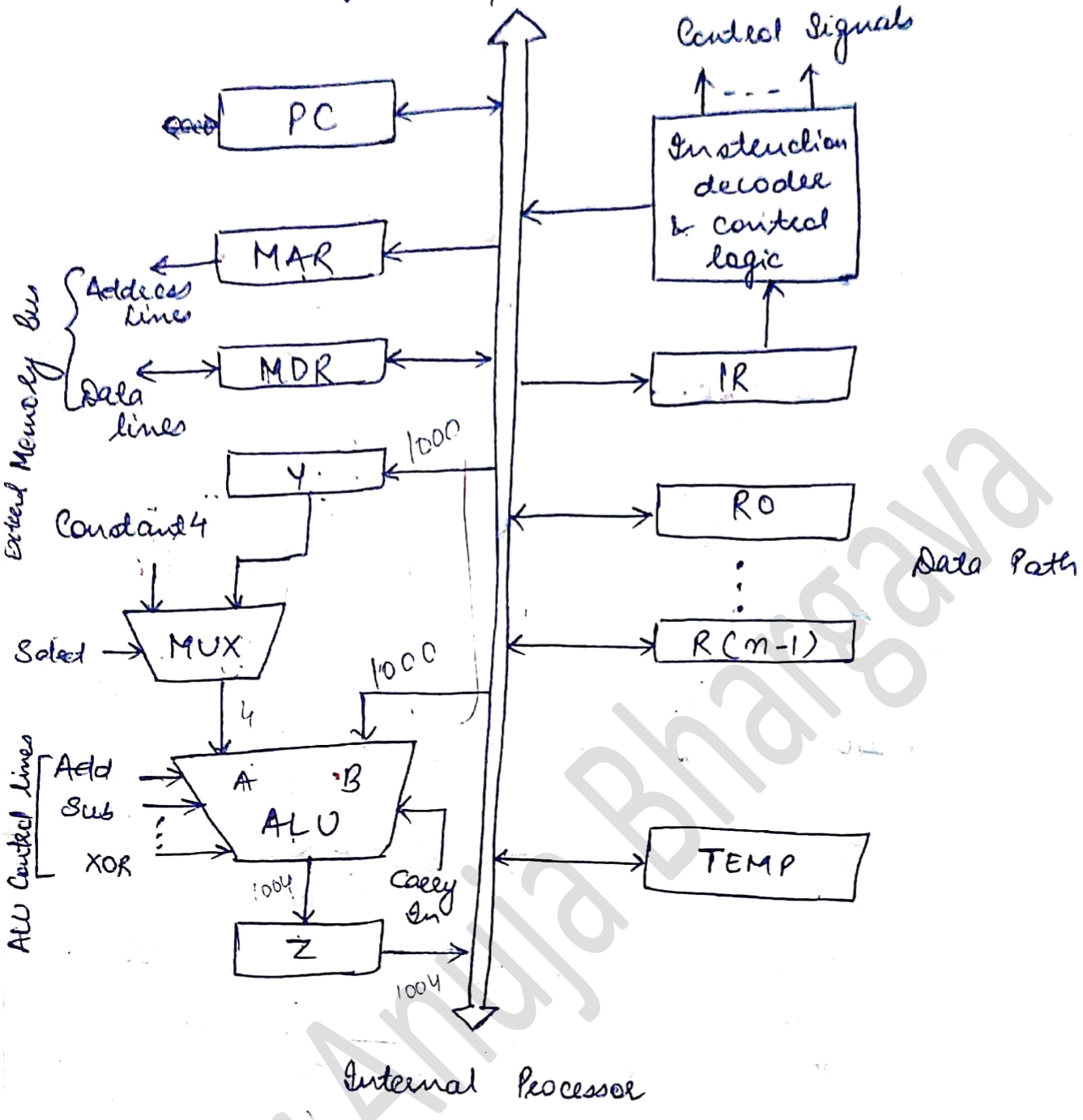


Fig:- Architecture of single bus processor

To execute the instruction, through a single bus following steps are to be followed.

1. Fetch the data
2. decode
3. Extract the address
4. Execute

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In figure

① R_0 to R_{n-1} are general purpose registers

② IR - Instruction / Indexed Register

It holds the instruction currently being executed

③ Instruction decoder - tells about the instruction to be performed

for fetching :- (i) AR ← PC

④ Y, Z & TEMP are temporary registers used by processor during the execution of some instruction.

⑤ PC : contains the address of next instruction to be executed.

⑥ MAR - Memory Address Register

⑦ MDR - All the data stored in Memory Data Register

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Memory

Address	Content
1000	Add(R3),R1
1008	20

PC 1000

R3 1008

R1 10

IR Add (R3), R1

Steps 1) Fetch the instruction from memory

2) Fetch the first operand from memory.

3) Perform addition operation.

4) Load the result into R1.

Control Sequence

1) PC_{out} , MAR_{in} , Read , Select 4 , Add , Z_{in}

2) Z_{out} , PC_{in} , WMFC

(Wait for memory function to complete)

3) MDR_{out} , IR_{in}

} Fetching instruction in IR

4) $R3_{out}$, MAR_{in} , Read

5) $R1_{out}$, Y_{in} , WMFC

6) MDR_{out} , Select Y, Add , Z_{in}

7) Z_{out} , R_{in} , End.

Mux → 4 Select

A = 4

B = 1000

Z = 1004

PC → 1004 Instr

MDR_{out} → ADD(R3) , R1

→ Add (R3) , R1

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Q Write control sequence for following instruction.

(a) ADD (R3), R1

1. PC_{out}, MAR_{in}, Read, Select 4, Add, Z_{in}
2. Z_{out}, PC_{in}, Y_{in}, WMFC
3. MDR_{out}, IR_{in}
4. R_3 out, MAR_{in}, Read
5. R1 out, Y_{in}, WMFC
6. MDR out, Select Y, Add, Z_{in}
7. Z_{out}, R_{in}, End.

* WMFC - Wait for memory function to complete.

b) Add the (immediate) number NUM to register R1.

1. PC_{out}, MAR_{in}, Read, Select 4, Add, Z_{in}
2. Z_{out}, PC_{in}, Y_{in}, WMFC
3. MDR_{out}, IR_{in}
4. IR NUM out, Y_{in}
5. R_1 out, Select Y, Add, Z_{in}
6. Z_{out}, $R_{1\,in}$, End

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c) Add the contents of memory location NUM to register R1.

1. PC_{out} , MAR_{in} , Read, Select 4 , Add, Z_{in}

2. Z_{out} , PC_{in} , Y_{in} , WMFC

3. MDR_{out} , IR_{in}

4. IR NUM $_{out}$, MAR_{in} , Read

5. $R1_{out}$, Y_{in} , WMFC

6. MDR_{out} , Select Y, Add, Z_{in}

7. Z_{out} , $R1_{in}$, End.

d) Add the content of memory location whose address is at memory location NUM to register R1.

1. PC_{out} , MAR_{in} , Read , Select 4 , Add, Z_{in}

2. Z_{out} , PC_{in} , Y_{in} , WMFC

3. MDR_{out} , IR_{in}

4. IR NUM $_{out}$, MAR_{in} , Read

5. WMFC

6. MDR_{out} , MAR_{in} , Read

7. $R_{1 out}$, Y_{in} , WMFC

8. MDR_{out} , Select Y, Add, Z_{in}

9. Z_{out} , $R1_{in}$, End.

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Q What is the sequence of operations to add the content of register R1 to those of R2 and store the result in R3 ?

1. $R1_{out}$, Y_{in}
2. $R2_{out}$, Select Y, Add, Z_{in}
3. Z_{out} , $R3_{in}$

Q Fetching a word from memory

MOV (R1), R2

1. $R1_{out}$, MAR_{in} , Read
2. MDR_{in} , WMFC
3. MDR_{out} , $R2_{in}$

Q Storing a word from memory.

MOV R2, (R1)

1. $R1_{out}$, MAR_{in}
2. $R2_{out}$, MDR_{in} , Write
3. MDR_{outE} , WMFC

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